CABE ROBERTSON

Las Cruces, NM (willing to relocate) | 575-644-4703 | caberobertson@gmail.com | caberobertson.github.io

SUMMARY

Electrical Engineering student (Communications & Signal Processing) interested in hardware development and cutting-edge technology. Hands-on embedded, RF, sensor, and benchtop test experience across three internships and a signal-processing research assistantship. IEEE MYOSA 6.0 finalist presenting live hardware at IEEE SENSORS 2026 in Rotterdam.

EDUCATION

New Mexico State University | Las Cruces, NM — **Expected December 2027**

B.S. Electrical Engineering, Concentration: Communications & Signal Processing

TECHNICAL SKILLS

Embedded & Hardware: ESP32 firmware (C/C++), BLE, I2C/SPI, PWM/BLDC motor drive, ADC/DAC interfacing, signal conditioning, EAGLE PCB design (schematic capture and layout), VHDL, FPGA, soldering and rework

Signals & Sensors: 6-axis IMU sensor fusion, channel coding and bit-error-rate analysis, passive filtering, RF communication-jamming systems, EEG recording (NeuroPawn kit)

Programming: Python (NumPy, OpenCV), C, C++, MATLAB, Flask, Git

Lab & Design: Oscilloscope, spectrum analyzer, function generator, design of experiments, AutoCAD (DWG/DXF)

Languages: English (Native), Spanish (Native)

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern | Emerging Technology Ventures (ETV America), NM — **Jul 2026 - Present**

- Sole engineer standing up the company's unmanned systems production floor end to end: AutoCAD equipment layout, tooling and power specification, vendor qualification, and procurement, delivering a turnkey 7-person build area ready for staff move-in
- Authored the capital-equipment budget and 59-line-item BOM (~$168K), sourcing every component to Blue UAS / NDAA-compliant vendors and delivering it as an audit-ready QuickBooks Enterprise import
- Co-authored a 30-page NASA SBIR/STTR proposal (CCRPP) seeking a $2.5M matching award for an autonomous on-orbit inspection and repair system
- Selected by company leadership to pitch at the NASA Moon to Mars Summit (Huntsville, AL, Sept 2026) on licensing NASA's VIPIR robotic inspection payload into company flight hardware, competing for a $10,000 award

Electrical Engineering Intern | GILZ LLC, Remote — **Mar 2026 - May 2026**

- Reverse-engineered a commercial BLDC blower to characterize motor voltage/current ratings and PWM response, defining the drive specification for an air-filtration prototype
- Programmed an ESP32 PWM controller (C / Arduino) adding 3 user-selectable speeds to a unit that shipped with no speed control
- Ran a benchtop design-of-experiments sweeping duct and filter configurations against airflow, acoustic noise, temperature, and velocity, identifying the one that cut noise ~10 dB and raised airflow 20%

Research Assistant | Mitchell Coding Group, NMSU, Las Cruces, NM — **Aug 2025 - Feb 2026**

- Developed Python simulations of linear-algebra channel decoding, validating bit-error-rate performance across encoding schemes under noise
- Ran multi-hour experiments over contrasting input bit-strings to verify decoder reliability, automating statistics and publication-quality plotting in MATLAB
- Worked independently on an assigned research question over seven months, reporting results to the principal investigator

Hardware Engineering Intern | Syndetix Incorporated, Las Cruces, NM — **Apr 2025 - Aug 2025**

- Designed and built a portable $219 demonstration kit for an RF communication-jamming device shown to law enforcement and defense clients, ~56% cheaper than the $500 setup
- Engineered a stabilizing fix that eliminated tip-over failures, keeping the RF demo operational in windy outdoor conditions during live client demos
- Authored the product user manual, standardizing setup, operation, and RF handling procedures for mission and training use

TECHNICAL PROJECTS

CrashGuard: Autonomous Crash Detection & Emergency Dispatch | Team Lead, 4-person team — **IEEE MYOSA 6.0 Finalist**

- Selected as 1 of 15 teams worldwide in the IEEE Sensors Council's international MYOSA 6.0 competition, then 1 of 7 advanced to the final round, presenting a live hardware demonstration at IEEE SENSORS 2026 (Rotterdam, October 2026)
- Own the ESP32/MPU6050 sensor-fusion firmware, flagging impacts from 6-axis IMU data at a 50 g threshold and autonomously dispatching an emergency call; tuned thresholds and GPS filtering to eliminate false triggers
- Integrated the detection-to-dispatch chain across BLE bridging, a Flask call server, and a Twilio voice interface, resolving threading and timeout failures that blocked reliable calls
- **EEG Signal Analysis | NeuroPawn Biopotential Kit:** Recorded and analyzed own EEG signals; hands-on with electrode placement, artifacts, and raw biopotential data
- **Real-Time Face Tracking | Python, OpenCV:** Built a real-time computer-vision tracking pipeline sustaining ~30 FPS on constrained hardware

LEADERSHIP & RECOGNITION

Vice President | IEEE NMSU Student Chapter — **Oct 2024 - Present**

- Tripled active membership from 10 to 30+; led the chapter turnaround that won NMSU's Most Improved Student Organization Award
- Lead a $50,000 fundraising initiative, serving as primary liaison to industry sponsors and national IEEE
- Run hands-on soldering and troubleshooting workshops, mentoring members through lab and prototyping skills